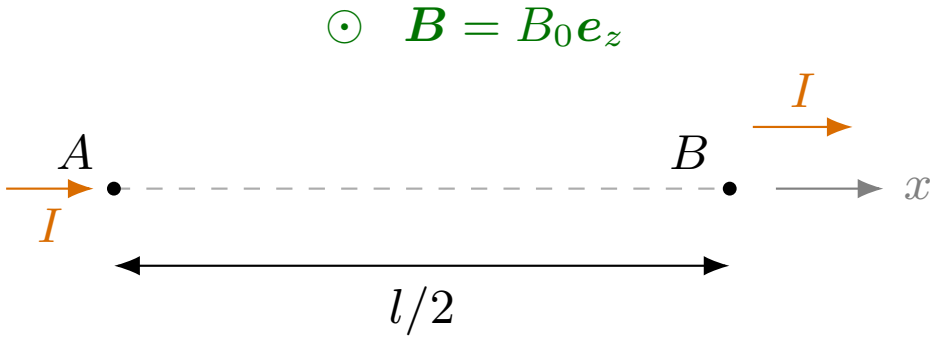
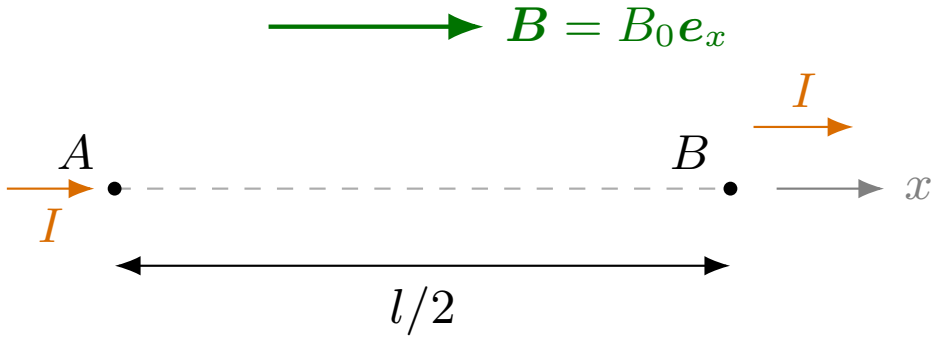
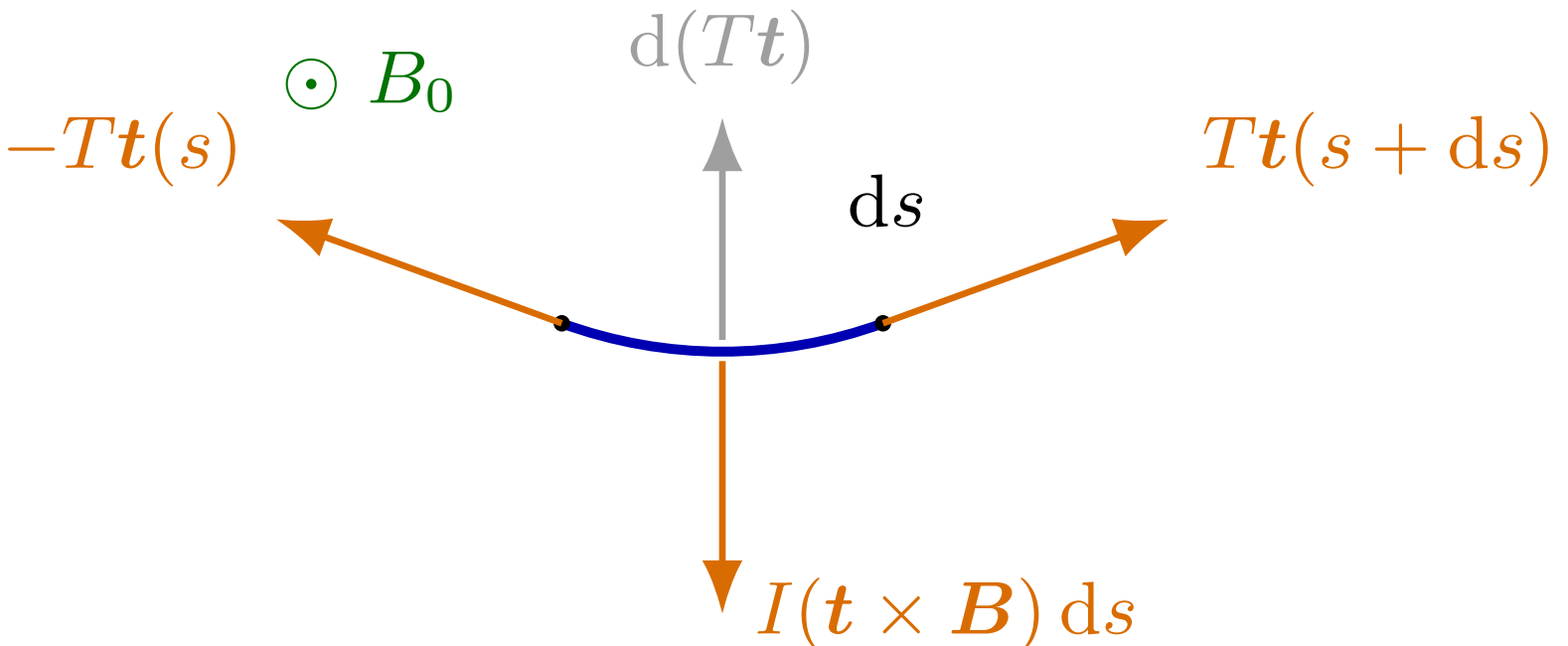
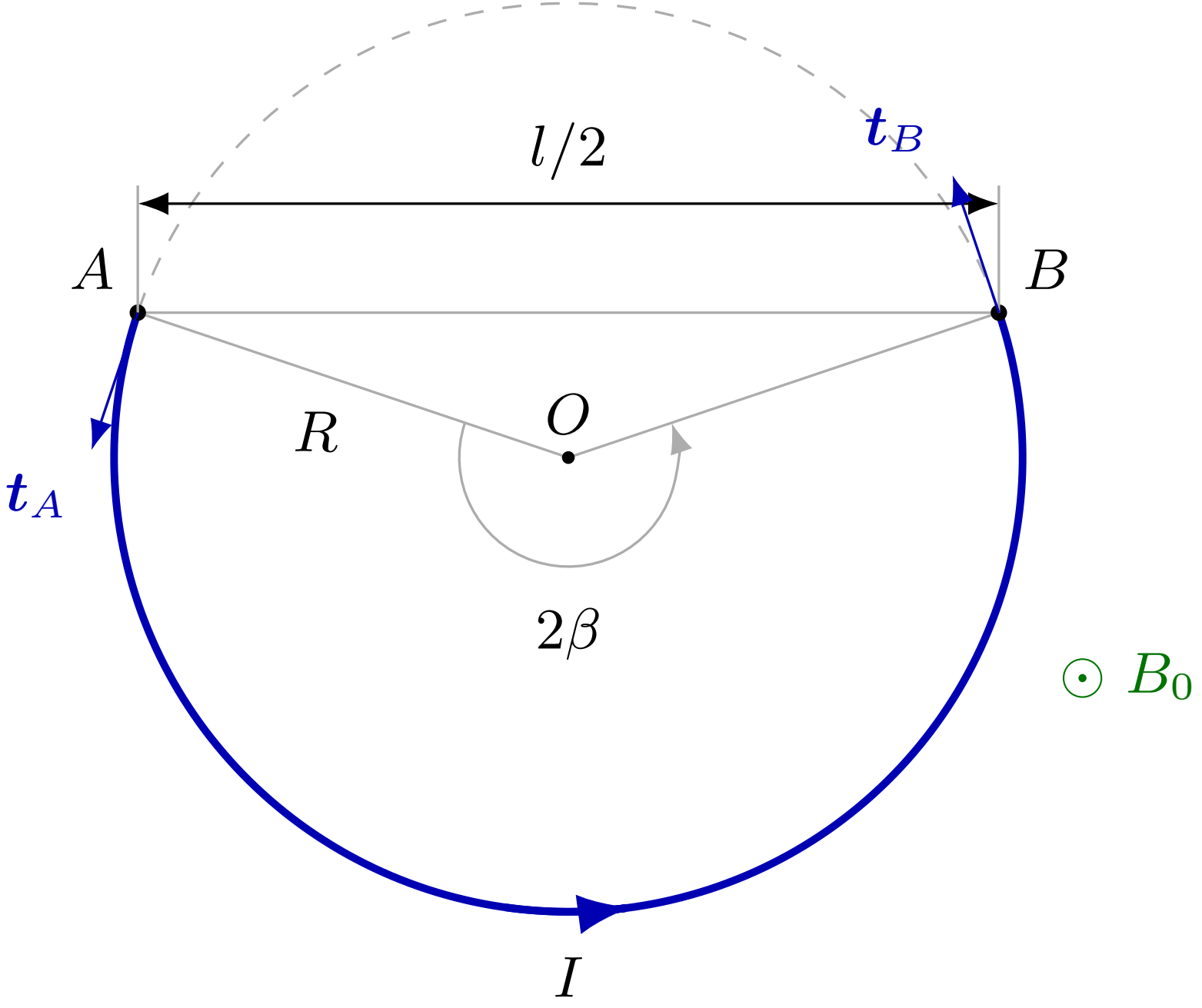
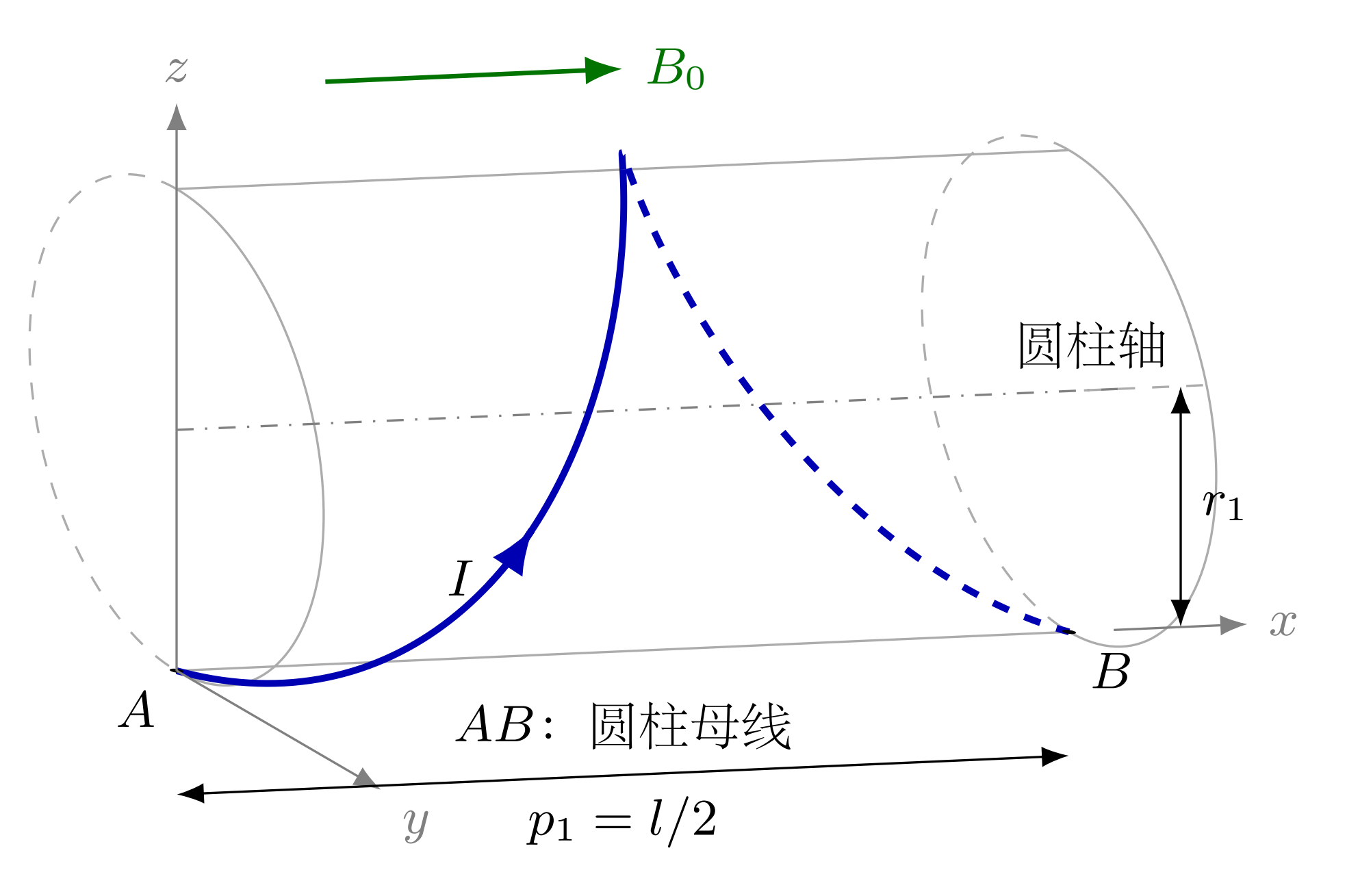
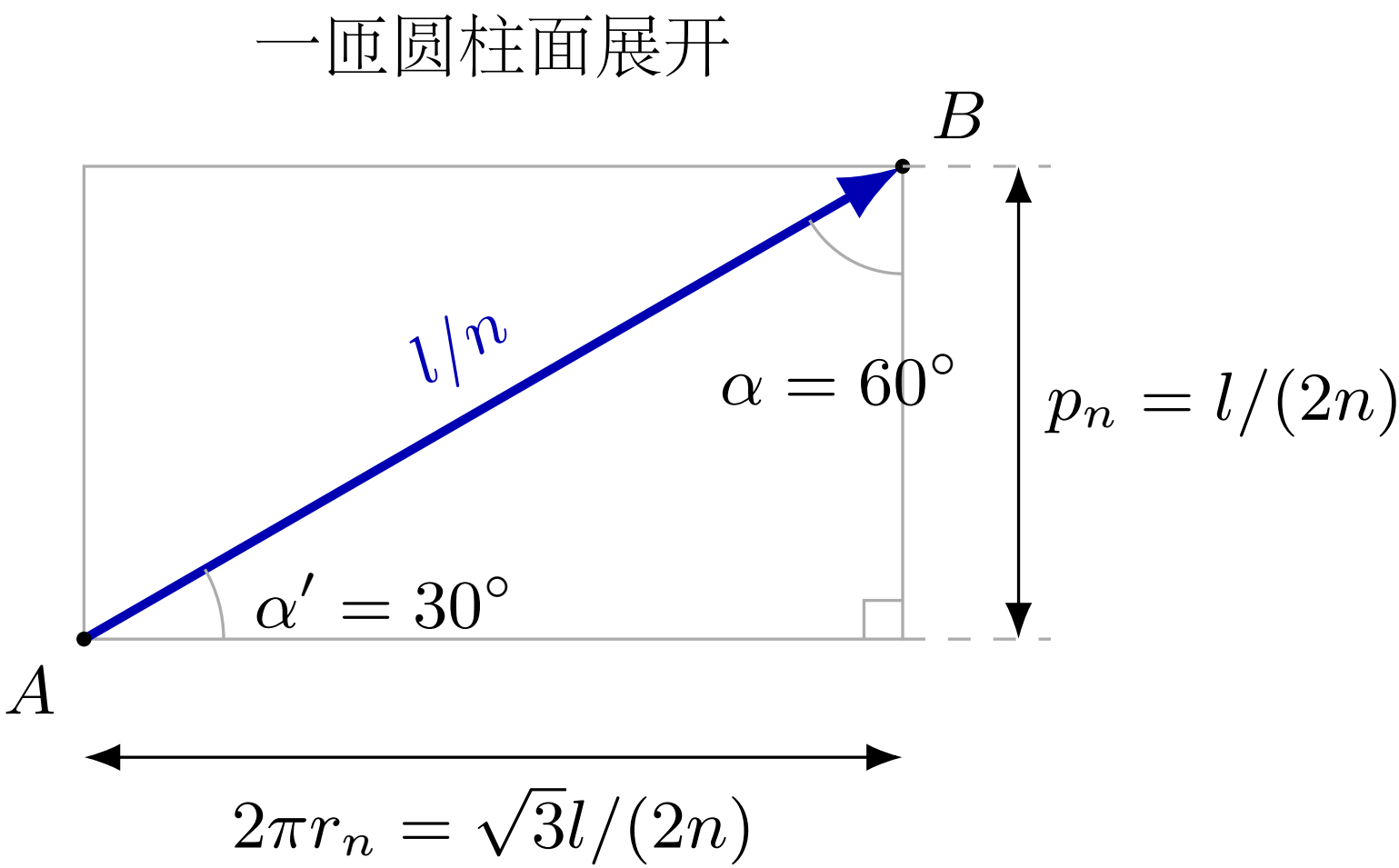

$\boldsymbol{B} = B_0 \boldsymbol{e}_z$
$I$
$A$
$B$
$x$
$I$
$l/2$


$\boldsymbol{B} = B_0 \boldsymbol{e}_x$
$I$
$A$
$B$
$x$
$I$
$l/2$


$\odot\ B_0$
$\mathrm{d}(T\boldsymbol{t})$
$-T\boldsymbol{t}(s)$
$T\boldsymbol{t}(s+\mathrm{d}s)$
$\mathrm{d}s$
$I(\boldsymbol{t}\times\boldsymbol{B})\,\mathrm{d}s$


$l/2$
$\boldsymbol{t}_B$
$A$
$B$
$R$
$O$
$\boldsymbol{t}_A$
$2\beta$
$\odot\ B_0$
$I$


z
$B_0$
圆柱轴
$r_1$
I
x
A
B
$AB$：圆柱母线
y
$p_1 = l/2$


一匝圆柱面展开
B
A
l/n
α = 60°
α′ = 30°
p_n = l/(2n)
2πr_n = √3 l/(2n)